Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
AI-500 and AI-501
Fall 2024
Mid Exam – Part I

Time Allowed: 1 hour and 40 minutes **Total Points**: 90 (Weightage: 40%)

Name: _____

Campus ID: _____

Problem 1. Select ALL (up to three) correct choices. Every incorrect answer would earn a penalty of 1 point but the total marks of any multiple-choice question will not be less than zero.

(1) Which of the following are valid subspace properties?

(a) Non-emptiness
(b) Closure under vector addition
(c) Dimension must be greater than or equal to 2
(d) Closure under scalar multiplication

(2) Which statements about orthonormal vectors are correct?

(a) Their dot product with each other is 1
(b) They have a norm of 1
(c) They are linearly dependent
(d) They can serve as a basis for a subspace

(3) Which of the following operations change the magnitude of a vector?

(a) Scalar multiplication
(b) Vector addition
(c) Dot product with another vector
(d) Normalization

(4) Which of the following are true about the Gram-Schmidt orthogonalization process?

(a) It creates an orthogonal set from linearly independent vectors
(b) It changes the dimensions of the original vectors
(c) It includes a normalization step
(d) It only applies to vectors in 2D spaces

(5) Our regression model is giving large variance error for different training data-sets. Which of the following would you prefer to use to reduce the variance?

(a) Add more data points to each training data-set.
(b) Change the optimization algorithm that is minimizing the convex loss function.
(c) Use model with lower complexity.

(d) Reduce the noise in each training data-set.

(6) Bias-variance trade-off is controlled or explained by
   (a) Regularization coefficient λ in L_2 regression
   (b) The learning rate α in the gradient descent
   (c) The polynomial degree in polynomial regression
   (d) The choice of activation function in neural network

(7) The computational complexity of predicting the label of a single test data point using the kNN algorithm depends on:
   (a) The size of validation data
   (b) The size of training data
   (c) The number of features
   (d) None of the above since we are predicting for a single test data point

(8) Which of the following is/are true for k in kNN algorithm?
   (a) Increasing k increases variance
   (b) Increasing k decreases variance
   (c) Increasing k decreases bias
   (d) Increasing k increases bias

(9) Which regularization term promotes sparse weight vector
   (a) L_1 regularization
   (b) L_2 regularization
   (c) Neither L_1 regularization, nor L_2 regularization
   (d) Either L_1 regularization, or L_2 regularization

(10) Which of the following performance metrics answers the question: "When it's positive, how often does the model predict positive"?
   (a) Recall
   (b) Accuracy
   (c) Specificity
   (d) Precision

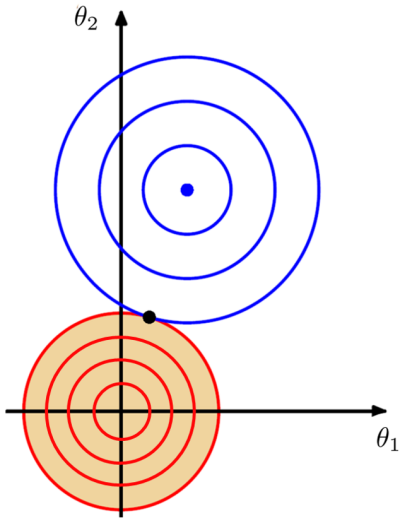
(11) For a binary classifier that predicts whether a customer will pay back the loan or not, the recall is 83%. Determine the correct implication of using the given classifier.
   (a) We can interpret that 83% of the all the customers will pay back.
   (b) 17% of the customers who could have paid back the loan will have their application rejected.
   (c) 83% of the customers who could have paid back the loan will have their application rejected.
   (d) We can interpret that 17% of the all the customers will pay back the loan.

(12) Which of the following is/are true about polynomial regression?
   (a) It is only applicable to data with a linear relationship.
   (b) It can capture non-linear relationships between the independent and dependent variables.
   (c) It involves fitting a polynomial curve to the data.
   (d) It can lead to overfitting if the degree of the polynomial is too high.

(13) Consider a classifier for the detection of cancer disease. True here refers to Cancer positive. We do not want to diagnose any cancer positive patient as healthy. Which of the following situations would you prefer for your classifier?

(a) FN $\gg$ FP
(b) FP $\gg$ FN
(c) FN = FP$\times$ TP
(d) TN $\gg$ FP

(14) For this question, consider the following figure depicting regularization during regression. We have a contour plot and and a constraint region for the representation of the error function (without regularization) and regularization term respectively. Choose the correct statement(s).



(a) This regularization is referred to as ridge regression.
(b) This regularization yields sparse solution.
(c) We have used L_1 penalty as regularization term.
(d) We have used L_2 penalty as regularization term.

(15) When a matrix $\mathbf{A}$ is symmetric, which of the following properties holds true?

(a) $\mathbf{A}^T = \mathbf{A}$
(b) It must be diagonal.
(c) Its eigenvalues are real.
(d) Its eigenvectors are orthogonal.

(16) Which of the following statements correctly describe the left-inverse of a matrix?

(a) The left-inverse exists only if the matrix has full row rank.
(b) It is used to solve overdetermined linear systems.
(c) A matrix can have multiple left-inverses. The left-inverse is always symmetric.
(d) A matrix can have multiple left-inverses.

(17) Suppose x and y are Boolean feature vectors, i.e., each entry is either 0 or 1, that encode the correct answers in the exam by you and your nearest neighbor in the exam hall. Which of the following are true statements?

(a) $x^T y$ is the number of correct answers you and your neighbor have in common.
(b) $\|x\|^2$ is the number of correct answers you have.
(c) $\mathbf{1}^T y$ is the number of correct answers your neighbor has.
(d) $\mathbf{1}^T(x - y)$ is the number of correct answers you have but your neighbor does not.

(18) If $Ax = 0$ for non-zero vector x, we can conclude that
(a) A=0
(b) The columns of A are linearly dependent
(c) The rows of A are linearly dependent
(d) All of the above

(19) What is the Frobenius norm of the following matrix A?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

(a) $\sqrt{15}$
(b) $\sqrt{30}$
(c) 5.47
(d) $\sqrt{25}$

(20) If a matrix $\mathbf{A}$ has an eigenvalue $\lambda = 0$, what can be concluded?
(a) The matrix is invertible.
(b) The matrix has linearly dependent columns.
(c) The determinant of the matrix is zero.
(d) None of the above.

(21) A matrix A has eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 5$. What are the eigenvalues of $3A^3 + 2I$, where I is the identity matrix?
(a) $\lambda_1 = 26$, $\lambda_2 = 127$
(b) $\lambda_1 = 28$, $\lambda_2 = 128$
(c) $\lambda_1 = 30$, $\lambda_2 = 130$
(d) $\lambda_1 = 24$, $\lambda_2 = 125$

(22) Consider a symmetric matrix $\mathbf{A}$. Which of the following conditions guarantee that $\mathbf{A}$ is positive semi-definite (PSD)?
(a) All eigenvalues of $\mathbf{A}$ are non-negative.
(b) $\mathbf{x}^T\mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
(c) $\mathbf{A} = \mathbf{B}^T\mathbf{B}$ for some matrix $\mathbf{B}$.
(d) $\det(\mathbf{A}) > 0$.

(23) Let $\mathbf{A}$ be a square matrix with full rank. Which of the following statements must hold for its **pseudo-inverse** $\mathbf{A}^+$?
(a) $\mathbf{A}^+ = \mathbf{A}^{-1}$ if $\mathbf{A}$ is invertible.
(b) $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$.
(c) $\mathbf{A}^+\mathbf{A} = I$.

(d) $\mathbf{A}^+$ always exists, even if $\mathbf{A}$ is not square.

(24) Let $\mathbf{A}$ be a 3×3 matrix with singular values $\sigma_1 = 5$, $\sigma_2 = 3$, and $\sigma_3 = 1$. What is the **determinant** of $\mathbf{A}^T\mathbf{A}$?

(a) 45
(b) 75
(c) 225
(d) 27

(25) For a vector x, we have $\|x\| = -\| - x\|$

(a) Always
(b) Sometimes
(c) Never